

1 PINN Method

1.1 Neural Networks

The Artificial Neural Network (ANN) is a computational approach to modeling the biological neural network's structure and functionality. ANNs are made up of connected units known as neurons, and these are organized in layers: there is an input layer, one or several hidden layers, and an output layer. A weight is assigned to each link, defining the strength of the information transmission across it.

In feedforward networks, information travels through successive layers from input to output, passing first through hidden layers. In each node, information enters and is processed, undergoing weighted addition and then nonlinear transformation with the use of an activation function. The training process involves modifying the network's parameters according to a loss function using algorithms for gradient descent such as backpropagation.

Neural networks are universal function approximators, meaning that, given enough training samples, they can approximate any continuous function. As a result of their capacity to discover intricate non-linear links between inputs and outputs, neural networks have found numerous applications in a variety of fields like image processing, text recognition, scientific computation, and the resolution of differential equations. In the context of Physics-Informed Neural Networks (PINNs), neural networks are employed to approximate the solution of a differential equation while simultaneously satisfying the governing physical laws through additional constraints incorporated into the loss function.[1]

Mathematically, a feed-forward neural network can be understood as a composition of several functions. Each layer transforms the output of the previous layer until the final output is produced. If the network has (L) layers, then the approximation can be written as

$$F_L \circ F_{L-1} \circ \dots \circ F_1(t, x)$$

where each (F_i) represents the transformation performed by the (i) -th layer. This layered structure allows the network to build increasingly complex representations of the solution.

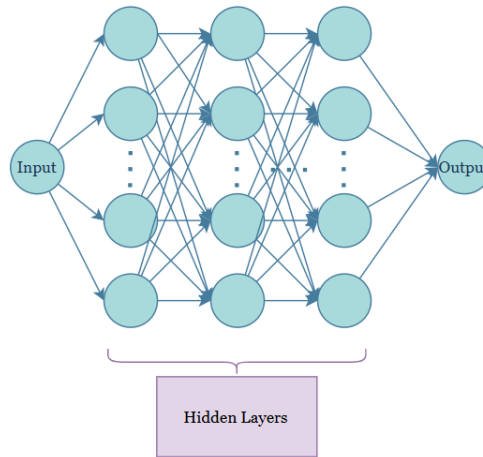


Figure 1: Neural Network

1.2 PINNs

The conventional neural network is trained with the help of data only. If we provide some data samples to the network, the network tries to minimize the error between the predicted output values and the observed values. Though this methodology has produced very good results in many different scenarios, it needs plenty of data and does not ensure that the solution that is predicted by the model complies with the physical laws.

On the other hand, Physics-Informed Neural Networks (PINNs) combine the concept of a neural network with the physical laws governing the system. Unlike a standard neural network, which is trained using data only, a PINN is constrained by physical laws represented by a Partial Differential Equation (PDE). Here, the neural network is responsible for approximating the unknown solution ($u(t, x)$), whereas automatic differentiation is used to calculate the necessary derivatives in the PDE. For a general PDE of the form

$$u_t + \mathcal{N}[u] = 0$$

the neural network predicts the solution ($u(t, x)$). Using automatic differentiation, the derivatives (u_t), (u_x), (u_{xx}), and higher-order terms can be computed exactly with respect to the network inputs. These derivatives are then substituted into the governing equation to form a residual function

$$f(t, x) = u_t + \mathcal{N}[u].$$

In case the neural network adheres exactly to the physical law, then the residual ($f(t, x)$) is supposed to take the value of zero throughout the whole domain. Hence, there will be two components in the training process of a PINN. One is related to the data error while the other component relates to physics, which serves as an additional constraint.

A PINN may therefore be regarded as a neural network that makes use of both observation data as well as the principles of physics. The incorporation of such physical knowledge provides additional information for the network, which works as a regularization technique.

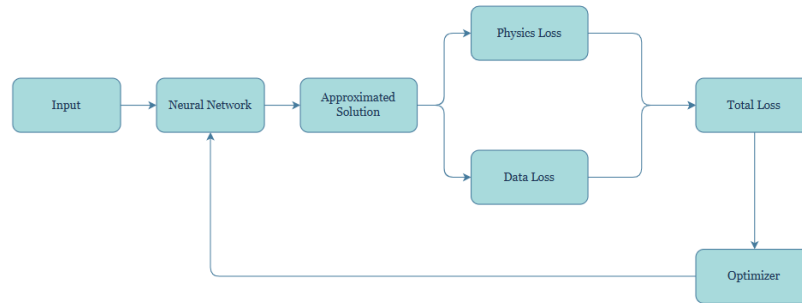


Figure 2: PINN Framework

1.3 Implenetation

The architecture of a Physics-Informed Neural Network is typically implemented based on a fully connected feed-forward neural network, though in many cases they try many other architecture for NN. In this structure, the input layer receives the independent variables of the problem. For a time-dependent one-dimensional PDE, these inputs are usually the spatial coordinate (x) and the time variable (t). Therefore, each input point can be shown as (x, t) . The role of the neural network is mapping this input point to the corresponding approximation of the unknown solution. In other words, the network represents the function

$$u_\theta(x, t) \approx u(x, t)$$

where (θ) denotes all trainable parameters of the neural network, including weights and biases[lulu] In the traditional neural network, the output of the architecture is checked only with available data. However, the output from the architecture in a PINN is further taken to the physics-based part of the algorithm. This means that after the network predicts $u_\theta(x, t)$, automatic differentiation is applied to compute the derivatives required in the PDE, such as

$$\frac{\partial u_\theta}{\partial t}, \frac{\partial u_\theta}{\partial x}, \frac{\partial^2 u_\theta}{\partial x^2}.$$

[raissi2019.pdf]

To evaluate how well the governing equation is satisfied, PINNs introduce a set of collocation points inside the computational domain. These points are sampled throughout the domain and are not necessarily associated with any observed data. Let the collocation points be denoted by

$(x_f^i, t_f^i)_{i=1}^{N_f}$ where (N_f) represents the number of collocation points. At each of these points, the residual $(f_\theta(x, t))$ is evaluated. If the residual is large at a particular location, it indicates that the neural network prediction violates the governing equation at that point. Consequently, the network parameters are adjusted to reduce this error. One important advantage of this approach is that PINNs are mesh-free methods and therefore do not require the computational mesh used by traditional numerical techniques such as the Finite Element Method or Finite Difference Method. [raissi2019] [Cuomo2022]

The learning process of a PINN is controlled through a loss function that combines information from both the available data and the governing physics. In the most general case, the total loss function consists of four components:

$$\mathcal{L}_{data} + \mathcal{L}_{physics} + \mathcal{L}_{boundary} + \mathcal{L}_{initial}$$

The data loss measures the difference between the neural network predictions and the available observations. The physics loss evaluates how well the governing differential equation is satisfied at the collocation points. The boundary loss enforces the prescribed boundary conditions, while the initial loss ensures that the initial conditions are satisfied. Together, these terms guide the network toward a physically meaningful solution.[raissi2019] More specifically, the physics loss is often defined as the mean squared value of the PDE residual evaluated at all collocation points:

$$\frac{1}{N_f} \sum_{i=1}^{N_f} |f_\theta(x_f^i, t_f^i)|^2.$$

Minimizing this term forces the neural network to satisfy the governing equation throughout the computational domain. Similarly, if observational data are available, the data loss can be written as

$$\frac{1}{N_u} \sum_{i=1}^{N_u} \left| u_{\theta}(x_u^i, t_u^i) - u^i \right|^2,$$

where (u^i) denotes the observed value of the solution at the point $((x_u^i, t_u^i))$. [raissi2019]
 After the total loss function is constructed, an optimization algorithm such as Adam or L-BFGS is used to update the trainable parameters (θ) . During each training iteration, the gradients of the loss function with respect to the network parameters are computed using backpropagation. The optimization process continues until the total loss reaches a sufficiently small value, indicating that the neural network simultaneously fits the available data and satisfies the governing physical laws. [Lu2021]